

AIML engineering student who ships production software and builds AI systems by hand from distributed backends to agent infrastructure

EXPERIENCE

SKF KARATE— Software Architect & Developer (Intern) | *Next.js + Supabase + TypeScript* [↗](#) Mar '26 – Aug '26

- Architected and **solo-built a production platform** for a national karate federation over a 6-month engagement, translating requirements gathered directly from students, coaches, and administrators into a system now serving thousands of monthly visitors.
- Designed a security-first data layer on Supabase/PostgreSQL with 63 Row-Level Security policies and atomic RPCs, eliminating race conditions in live tournament scoring and fee transactions.
- Shipped an installable PWA athlete portal with push notifications and an integrated fee-management module with automated receipts and offline-capable transaction logging.

PROJECTS

TERMINAL AGENT — Autonomous CLI Coding Assistant | *Python + OpenRouter API* [↗](#)

- Built an **autonomous terminal coding agent** entirely from scratch in Python with zero agent frameworks, implementing a think → act → observe tool-calling loop across 7 hand-coded tools spanning file read/write/edit, shell execution, and cross-file search.
- Designed a **tool-dispatcher architecture** that separates LLM reasoning from tool execution, letting the agent complete multi-step tasks through direct OpenRouter integration without manual step-by-step supervision.
- Maintained **full conversational and tool-call history** within each session, enabling the agent to reason over prior tool outputs and continue complex tasks without losing context mid-run.

RETAIN — Persistent Memory Layer for AI Agents (In Design) | *Python + SQLite* [↗](#)

- Designing a **hand-coded TF-IDF retrieval engine** over SQLite — tokenization, term frequency, inverse document frequency, and cosine similarity — as a lightweight, fully local alternative to embedding-based RAG.
- Modeling memory strength as an **Ebbinghaus-curve decay function**, so frequently retrieved context stays weighted high while unused context fades over time.

MODULAR ROUTING— Cost-Aware LLM Model Selection (*In Progress*) | *Python + OpenRouter API + Heuristics* [↗](#)

- Building a **complexity-aware LLM routing layer** that automatically directs simple queries to cheap, fast models and complex queries to more capable ones — aimed at the inference-cost problem behind incidents like Uber's 2025 rollback of internal AI tooling access.
- Developing **query-complexity classification** and **cross-model cost tracking** as the core routing logic, with usage-scoping (work vs. personal use) planned as a next layer.

CENTRALIZED EXCHANGE — Distributed Trading Backend | *TypeScript + Express + Redis* [↗](#)

- Architected a **distributed exchange backend** where a REST API and a separate matching-engine process communicate asynchronously through Redis message queues, using correlation IDs to route each engine response back to its pending HTTP request.
- Modeled the **exchange's relational schema** — Users, Stocks, Orders, and Fills — in PostgreSQL via Prisma 7, with full order-lifecycle tracking from open through partially-filled to filled or cancelled.
- Enabled the **API layer to scale independently** of the matching engine — each backend instance registers its own Redis response queue and a correlation-ID-keyed pending-request store with timeout handling for delayed engine replies.

MERCHANDISE — Full-Stack P2P Payments Platform | *React + Express + MongoDB* [↗](#)

- Built a full-stack peer-to-peer payments platform (React 19/TypeScript frontend, Express/MongoDB backend) with **atomic wallet transfers** executed inside MongoDB session-based transactions, guaranteeing all-or-nothing settlement on every transfer.
- Enforced **end-to-end transaction safety** — balance verification, account validation, and automatic rollback on failure — mirroring the consistency guarantees production financial systems rely on.
- Implemented **JWT authentication and bcrypt password hashing** from scratch alongside Zod-validated APIs and a complete, queryable transaction history per account.

TECHNICAL SKILLS

- **Languages:** TypeScript, Python, JavaScript, Java, SQL
- **Frontend:** React, Next.js, Tailwind CSS
- **Backend:** Node.js, Express.js, Bun, Supabase, REST APIs
- **Databases:** PostgreSQL, MongoDB, Redis, SQLite, Prisma
- **AI & Agents:** LLM Integration, Agentic Workflows, Memory Systems, Tool Calling, OpenRouter
- **Concepts:** System Design, Authentication (JWT, bcrypt), Real-Time Systems
- **Tools:** Git, GitHub, Zod, Cloudinary, Vercel

EDUCATION

Bachelor of Engineering / **Artificial Intelligence & Machine Learning**
East West Institute of Technology, Bangalore

Aug '23 – Present
CGPA: 8.0

LEADERSHIP & ACTIVITIES

National-Level Karate Champion | SKF Karate – Gold & Bronze Medalist

Karate Coach | SKF Karate – **Instructor** to **100+ students**

Freelancer | Rocksport – Organized outdoor activities, guided groups, built teamwork & leadership

[Participant](#) | Lazarus Mission, IEEE NITK Surathkal – **Hackathon**